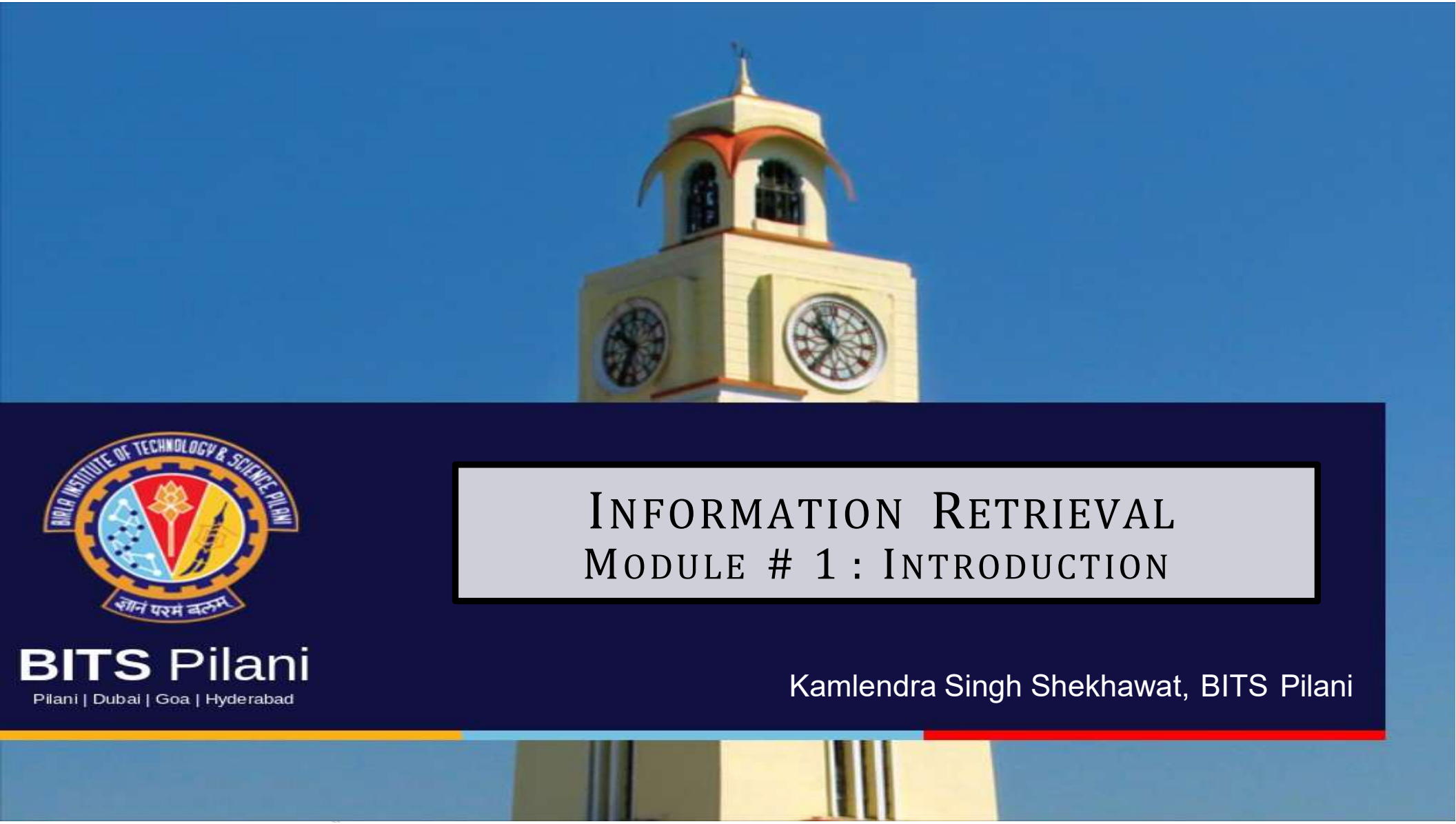

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INFORMATION RETRIEVAL
MODULE # 1 : INTRODUCTION

Kamlendra Singh Shekhawat, BITS Pilani

The instructor is gratefully acknowledging the authors who made their course materials freely available online.

This deck is prepared by Kamlendra Singh Shekhawat.

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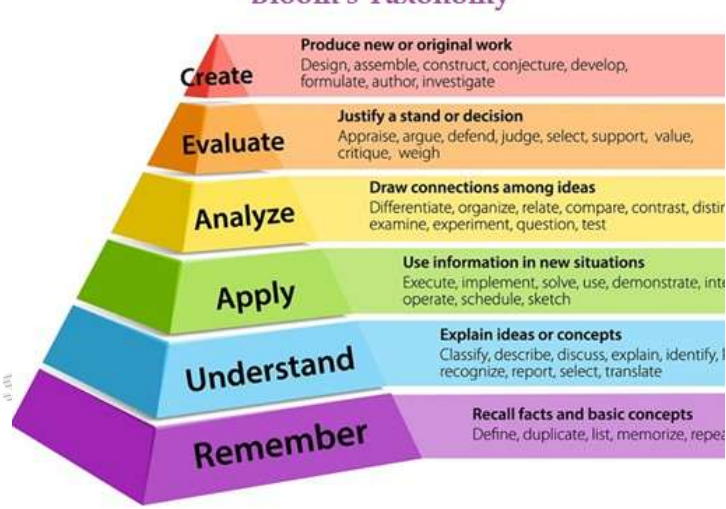
Course Objectives

CO1 To understand structure and organization of various components of an IR system.

CO2 To understand information representation models, term scoring mechanisms, etc. in the complete search system.

CO3 To understand architecture of search engines, crawlers and the web search.

CO4 To understand cross lingual retrieval and multimedia information retrieval.



Course Handout = What we Learn...

1. Basic Information Retrieval Concepts
2. Vector Space Model
3. Text Mining
4. Video Segmentation (6 hrs) (T1 Ch 6)
5. Evaluation in Information Retrieval
6. Web Search
7. Cross Lingual Retrieval
8. Multimedia Information Retrieval
9. Recommender Systems
10. Neural IR

Pre-requisites

1. Data structures
2. Graph theory
3. Probability theory
4. Programming experience is necessary for Assignments.

Text and Reference Books

Text Books

T1	C. D. Manning, P. Raghavan and H. Schutze. Introduction to Information Retrieval. 2008. http://nlp.stanford.edu/IR-book/

Reference Books

R1	Modern Information Retrieval, Ricardo Baeza-Yates and Berthier Ribeiro-Neto
R2	Ricci, F. Rokach, L. Shapira, B. Kantor, P.B., Recommender Systems Handbook, 2011.
R3	Cross-Language Information Retrieval by By Jian-Yun Nie, 2010
R4	Multimedia Information Retrieval by Stefan M. Rüger, 2010.
R7	An Introduction to Neural Information Retrieval, Bhaskar Mitra, Nick Craswell https://www.microsoft.com/en-us/research/uploads/prod/2017/06/fntir2018-neuralir-mitra.pdf

Demo / Lab Sessions

1. Demo on IR tools - Python and Java

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LMS

Most relevant and up to date info on Canvas

- Handout
- Schedule for Demo, Quiz, and Assignments.
- Session Slide Decks
- Demo Lab Sheets
- Quiz-I, II and III
- Assignment-I

Evaluation Schedule

No	Name	Type	Duration	Weight
EC1	Quiz I	Online	0.5 hr	5 %
	Quiz II	Online	0.5 hr	5 %
	Quiz III (best 2/3)	Online	0.5 hr	5 %
	Assignment I	Online	2 weeks	20 %
EC2	Mid-sem Regular	Closed book	2 hrs	30 %
EC3	Compre Regular	Open book	2.5 hrs	40 %

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Module Topics....

- Information Retrieval
- Information vs Data Retrieval
- Basic Search Model
- The retrieval process
- Taxonomy of IR
- Classic IR and Alternative models

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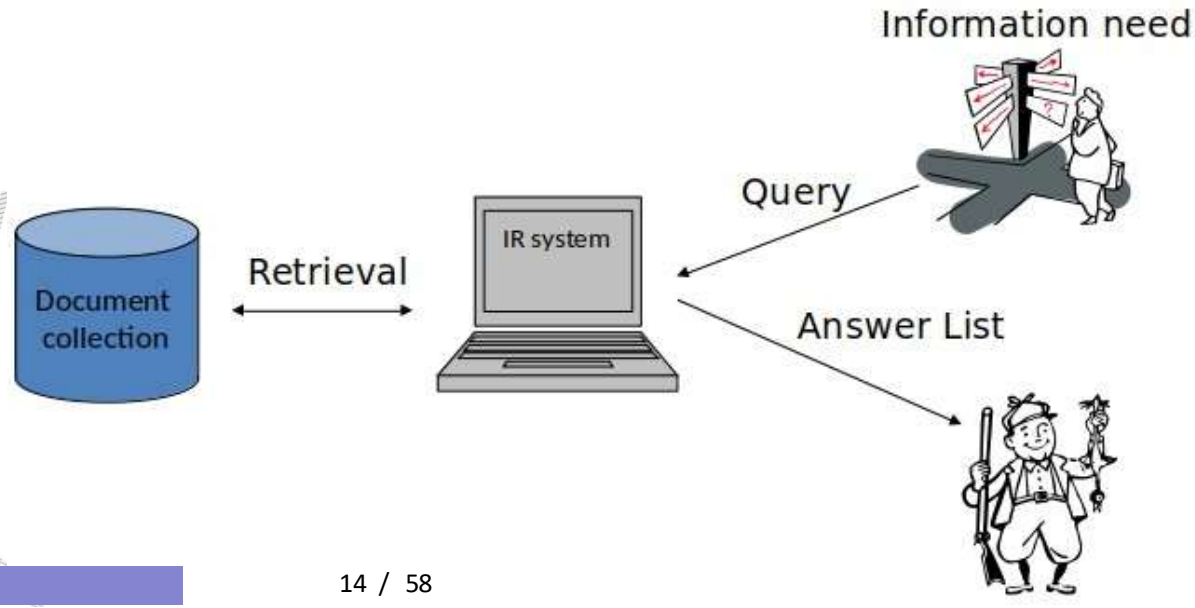
Information Retrieval

Definition

Information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers).



ler Section 3 of UGC Act, 1956

Information Retrieval

- Information retrieval every day when you use a web search engine or search your email.
- Information retrieval is fast becoming the dominant form of information access.
- IR is also used to facilitate semistructured search such as finding a document where the title contains Java and the body contains threading.

Unstructured Data

- Data which does not have clear, semantically overt, easy-for-a-computer structure. eg: text, image, XML, JSON, source code
- Opposite of structured data, eg: spreadsheet data, relational database, personnel records, product inventories

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Information Retrieval vs Data Retrieval

- Data retrieval
  - Which documents contain a set of keywords?
  - Well defined structure and semantics. Satisfy clearly defined conditions.
  - A single erroneous object implies failure.
  - Provide solution to the user of a database system.
- Information retrieval
  - Information about a subject or topic.
  - Semantics is frequently loose.
  - Small errors are tolerated.
  - Deals with natural language text, which can be semantically ambiguous.

Information Retrieval vs Data Retrieval

Characteristic	Data	Information
Document Property	Structured	Unstructured
Fields	Clear Semantics (SSN, age)	No fields (other than text)
Queries	Defined (relational algebra, SQL)	Boolean and Free text (natural language)
Matching	Exact (results are always correct)	Imprecise and approximate (need to measure effectiveness)
Results	Unordered	Ordered by relevance
Accessibility	Knowledgable users or automatic processes	non-expert humans

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Types of Information Need

- Retrospective
  - Searching the past
  - Different queries posed against a static collection
  - Time invariant
- Prospective (Filtering)
  - Searching the future
  - Static query posed against a dynamic collection
  - Time dependent

Retrospective Searches

- Adhoc retrieval: find documents about a topic.
  - Compile a list of endangered mammals.
- Known item search.
  - What is the ISBN of ”Modern Information Retrieval”.
- Directed exploration
  - What is the current version of ChatGPT?
- Question and answering in the form of factoid, list, and definition.
  - Who discovered oxygen?
  - What countries export oil?
  - Who is Roger Federer?

Prospective Searches

- Filtering: Make a binary decision about each incoming document
  - Email is a Spam or not?
- Routing: Sort incoming documents into groups.
  - Categorize new articles: world? nation? sports? finance? entertainment?

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Scale at which IR Operate

- Web Search
  - Search over billions of documents stored on millions of computers.
- Enterprise, institutional, and domain-specific search
  - Retrieval for collections such as a corporation's internal documents, a database of patents, or research articles.
- Personal information retrieval
  - Search in Email programs and text classification like spam filter.

Exercise

- Organise yourself in groups.
- Discuss with neighbours
  1. One example for each of scope of IR.
  2. How often do you search and retrieve documents per day on an average?
  3. Which type of information were you retrieving on an average?

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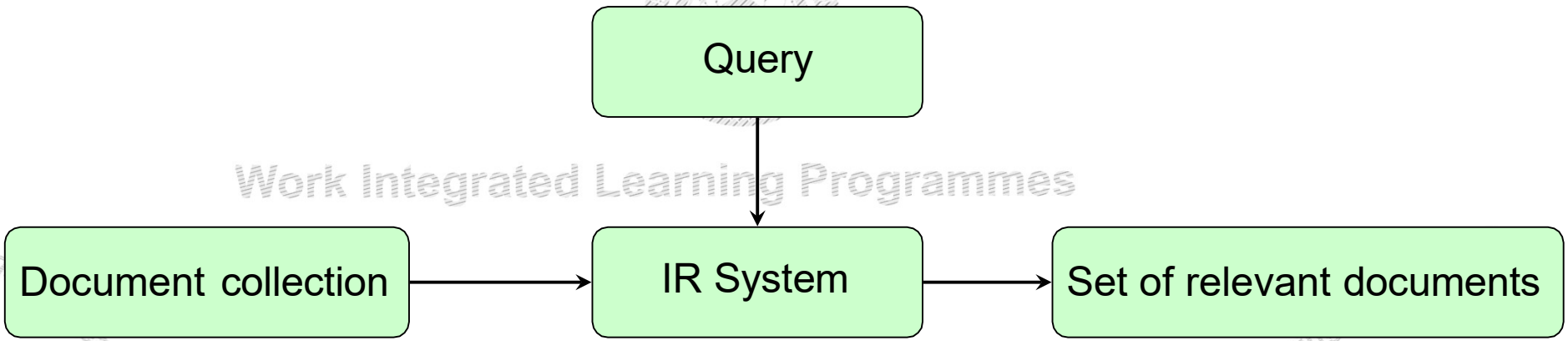
Basic Assumptions of IR

- Collection: Fixed set of documents
- Goal: Retrieve documents with information that is relevant to the user's information need and helps the user to complete a task.



RELEVANCE

- Relevance is a subjective judgment and may include
  - Being on the proper subject.
  - Being timely (recent information).
  - Being authoritative (from a trusted source).
  - Satisfying the goals of the user and intended use of the information (information need).

Intelligent IR

- Meaning of the words used.
- Order of words in the query.
- Direct or indirect feedback.
- Authority of the source

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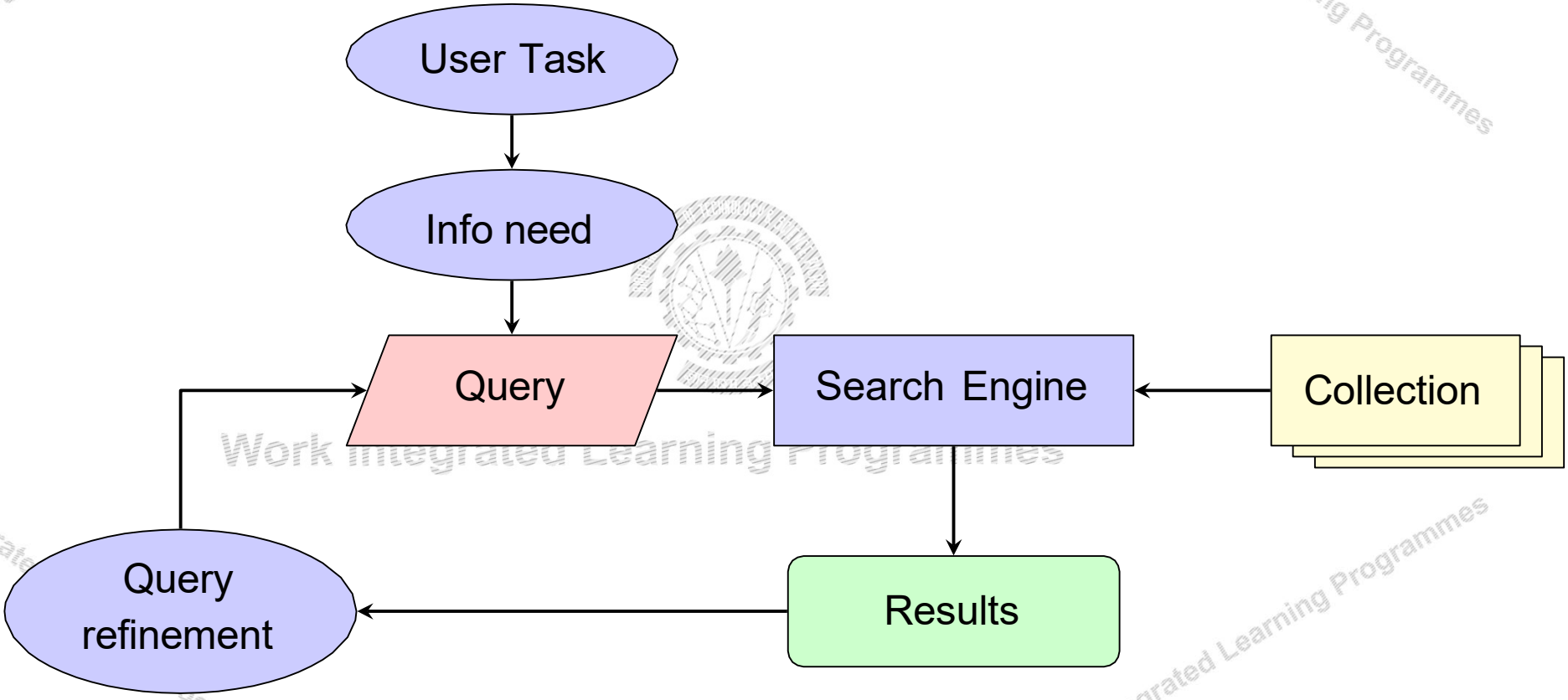
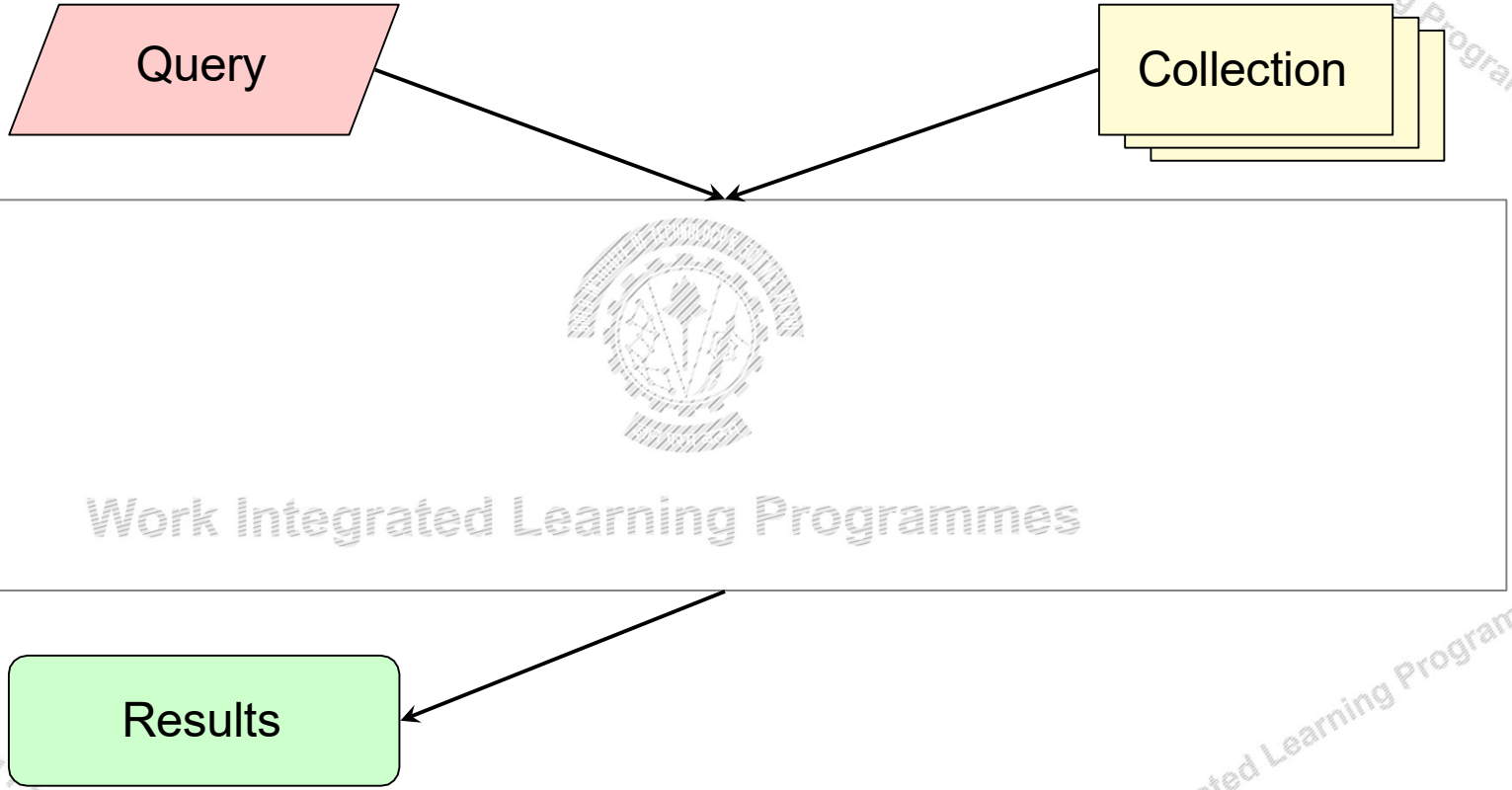
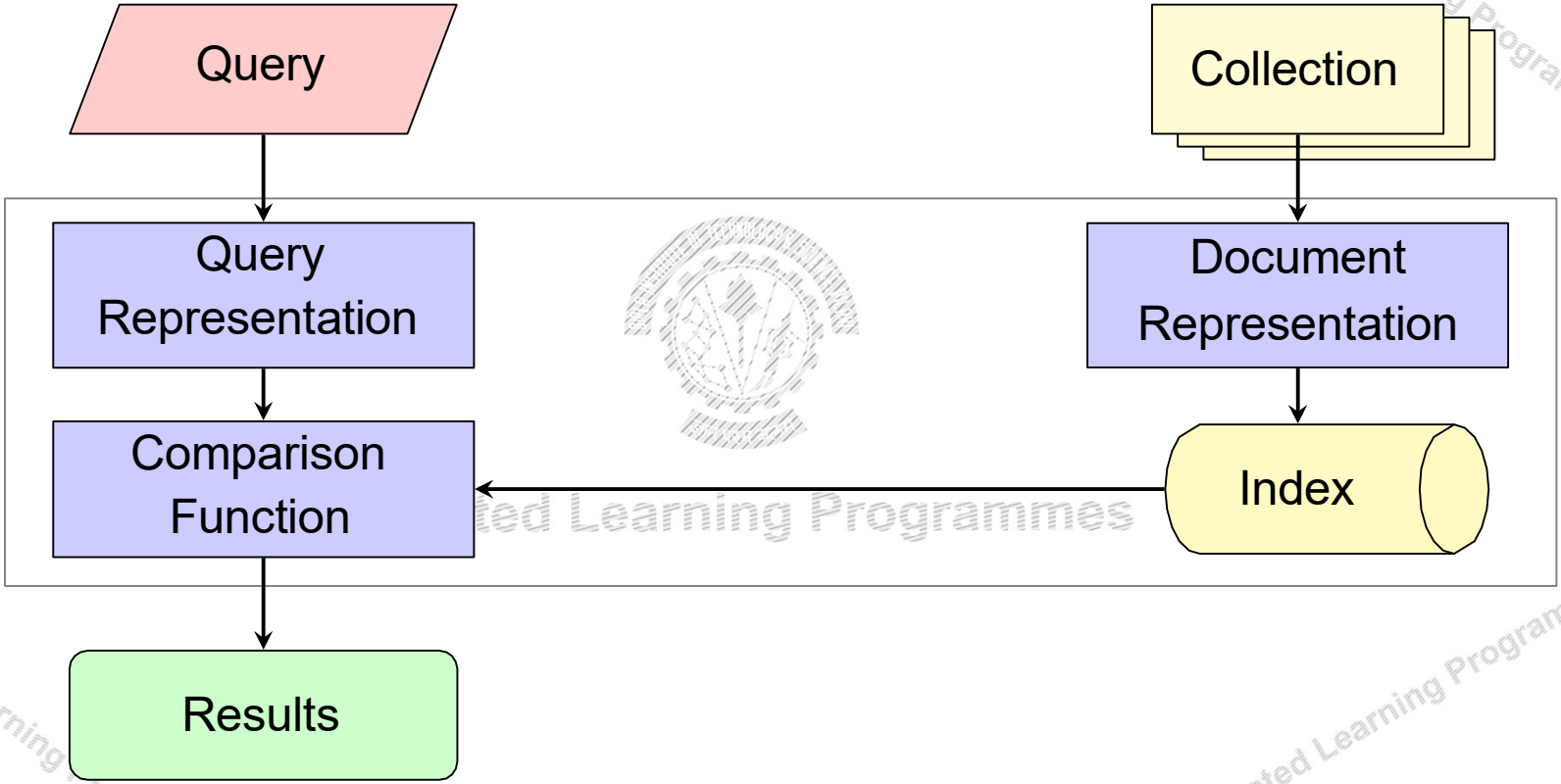
Classic Search Model



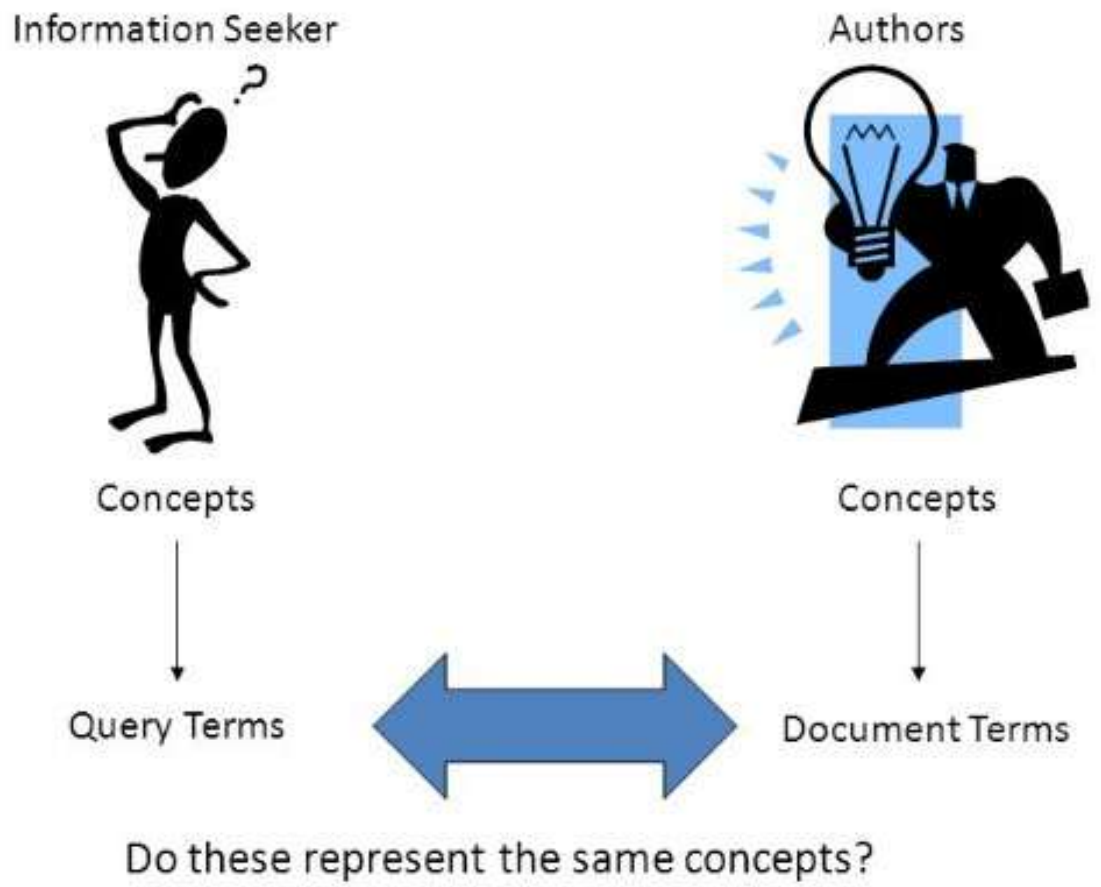
IR Black Box



IR Black Box


Query
Collection
Query Representation
Document Representation
Comparison Function
Index
Results

Central Problem in IR



Tasks of IR

- Index the documents in the collection (offline).
- Process the query
- Evaluate Similarity and find RANKs.
  - Find documents most closely matching the query (relevant documents).
- Display results.
- User may refine the query (feedback).

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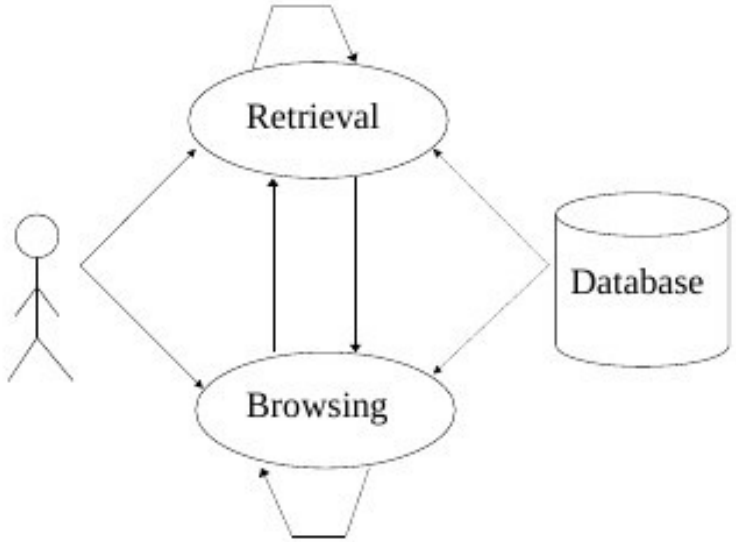
User Task

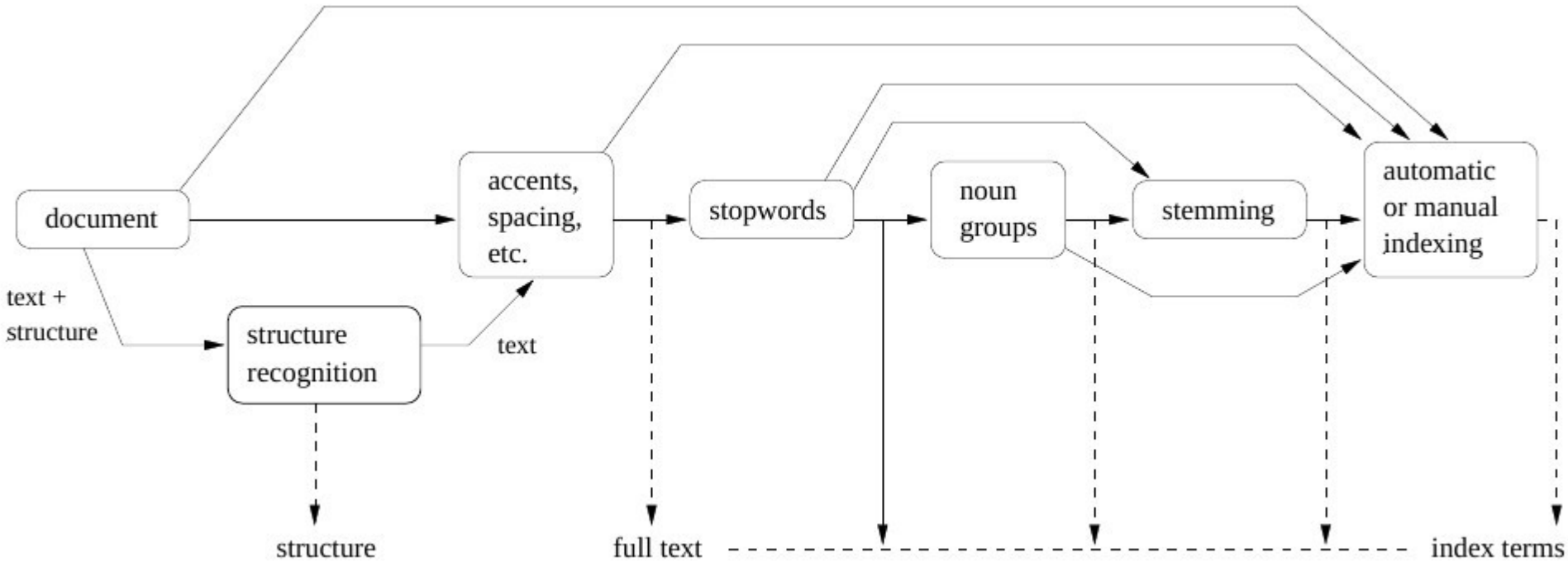
- Retrieval
  - Information or data
  - Purposeful
  - Translate the information need into a query in the language provided by the system.
  - Specify a set of words which convey the semantics of the information need.
- Browsing
  - Hypertext systems are used
  - Glancing around
  - Going through the documents in the collection and not searching.

User Task



- Both retrieval and browsing are pulling actions.
- Alternative is automatic to push the information towards the user, to execute the particular retrieval task which consists of filtering relevant information.

Logical View of Documents



Logical View of Documents

- Documents in a collection are frequently represented through a set of index terms or keywords.
- Keywords
  - Keywords are extracted from document.
  - Keywords are derived automatically or generated by a specialist, they provide a logical view of the document.
- Text operations – reduce the complexity of the document representation and allow moving the logical view from that of full text to that of a set of index terms.
  - Stop-words
  - Stemming
  - Noun Groups

Logical View of Documents

- Stop-words
  - To reduce the set of representative keywords from large collection.
  - Function words do not bear useful information for IR, i.e. of, in, about, with, I, although
  - Stop-list: contain stop-words, not to be used as index.
    - Prepositions, Articles, Pronouns
    - Some adverbs and adjectives, some frequent words (e.g. document)
    - A few standard stop-lists are commonly used.
  - The removal of stop-words usually improves IR effectiveness.

Logical View of Documents

- Noun groups
  - Identify the noun groups
  - Eliminate the adjectives, adverbs and verbs.
- Stemming
  - Different word forms may bear similar meaning, hence create a ”standard” representation for them.
  - Reduces distinct words to their common grammatical root.
  - Removing some endings of word.

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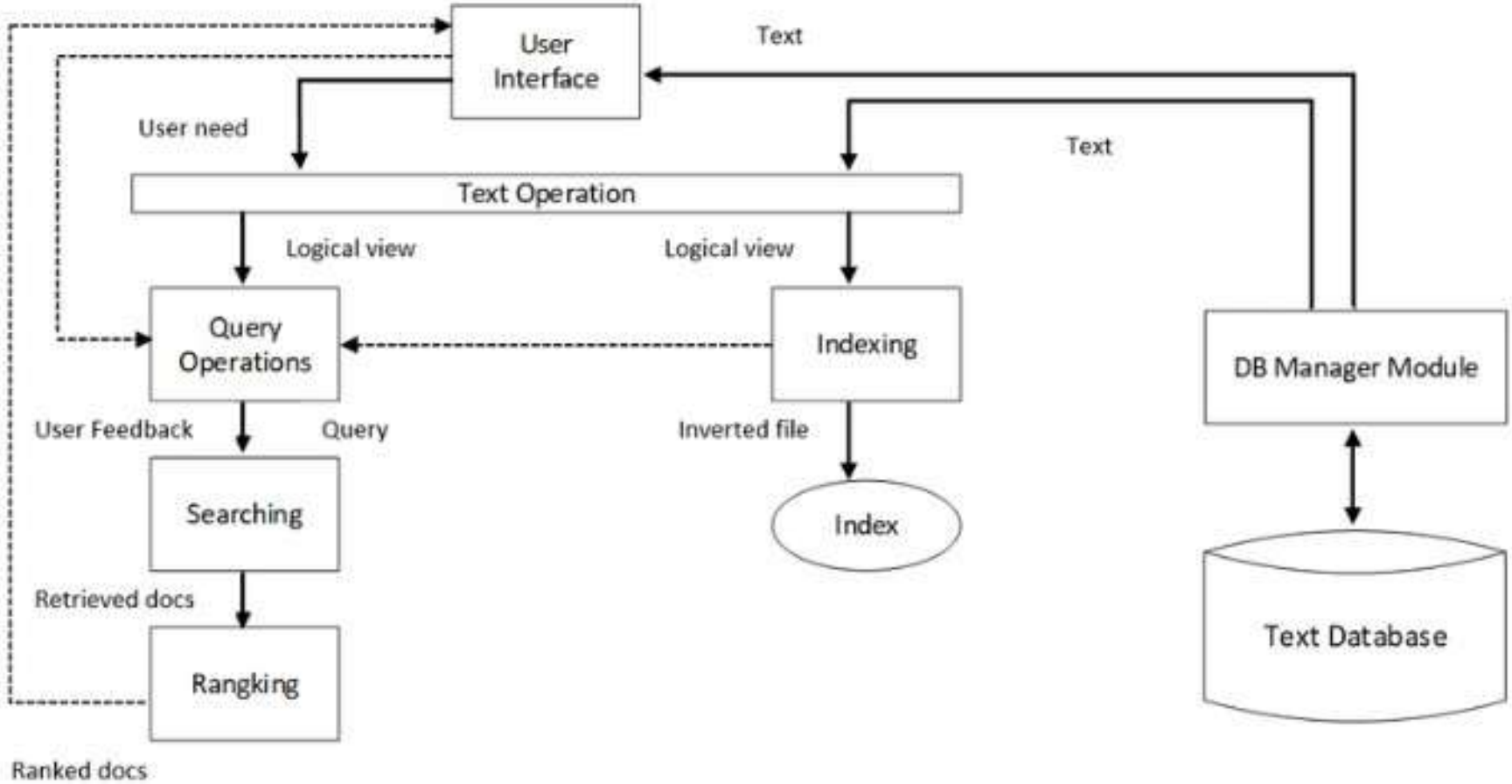
Process of Retrieving Information



Process of Retrieving Information

1. Define the text database. This is done the DB manager, which specifies
   - The documents to be used
   - The operations to be performed on the text. Text operations transform the original documents and generate a logical view of them.
   - The text model, i.e. the text structure and what elements can be retrieval
2. The database manager builds an index of the text i.e. inverted file. Index allows fast searching.
3. User need is parsed and translated by the same text operations.
4. Query operations are used to generate actual query based on the used needs.
5. Query is then processed to obtain the relevant documents.
6. The retrieved document ranked according to the likelihood of relevance.

IR System Components

1. Text Operations forms index words (tokens).
   - Stop-word removal
   - Stemming
2. Indexing constructs an inverted index of word to document pointers.
3. Searching retrieves documents that contain a given query token from the inverted index.
4. Ranking scores all retrieved documents according to a relevance metric.
5. User Interface manages interaction with the user
   - Query input and document output.
   - Relevance feedback.
   - Visualization of results.
6. Query Operations transform the query to improve retrieval
   - Query expansion
   - Query transformation using relevance feedback

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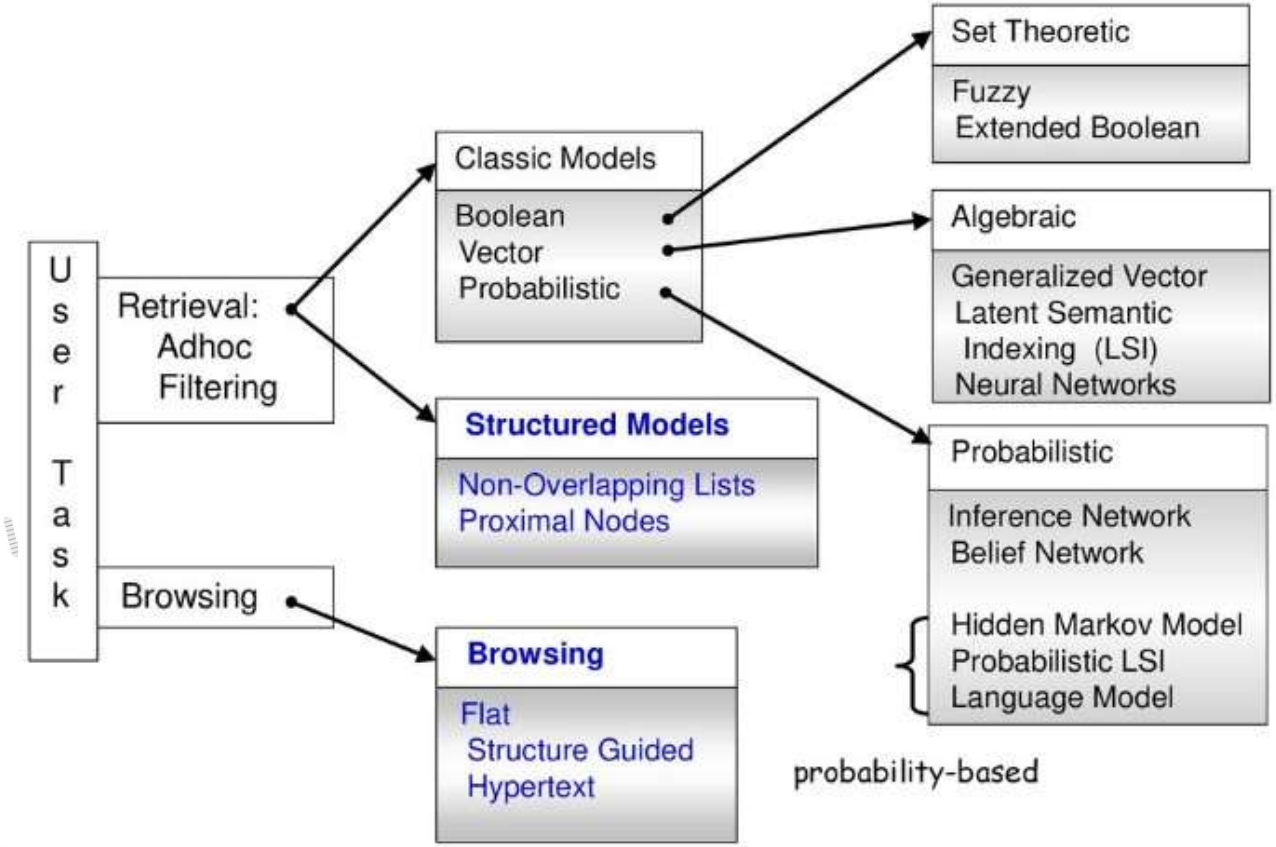
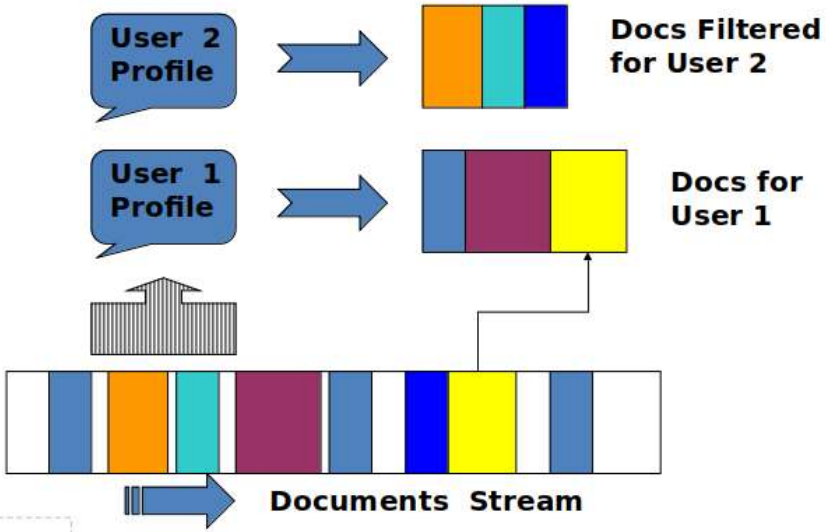
Taxonomy of IR – User Task



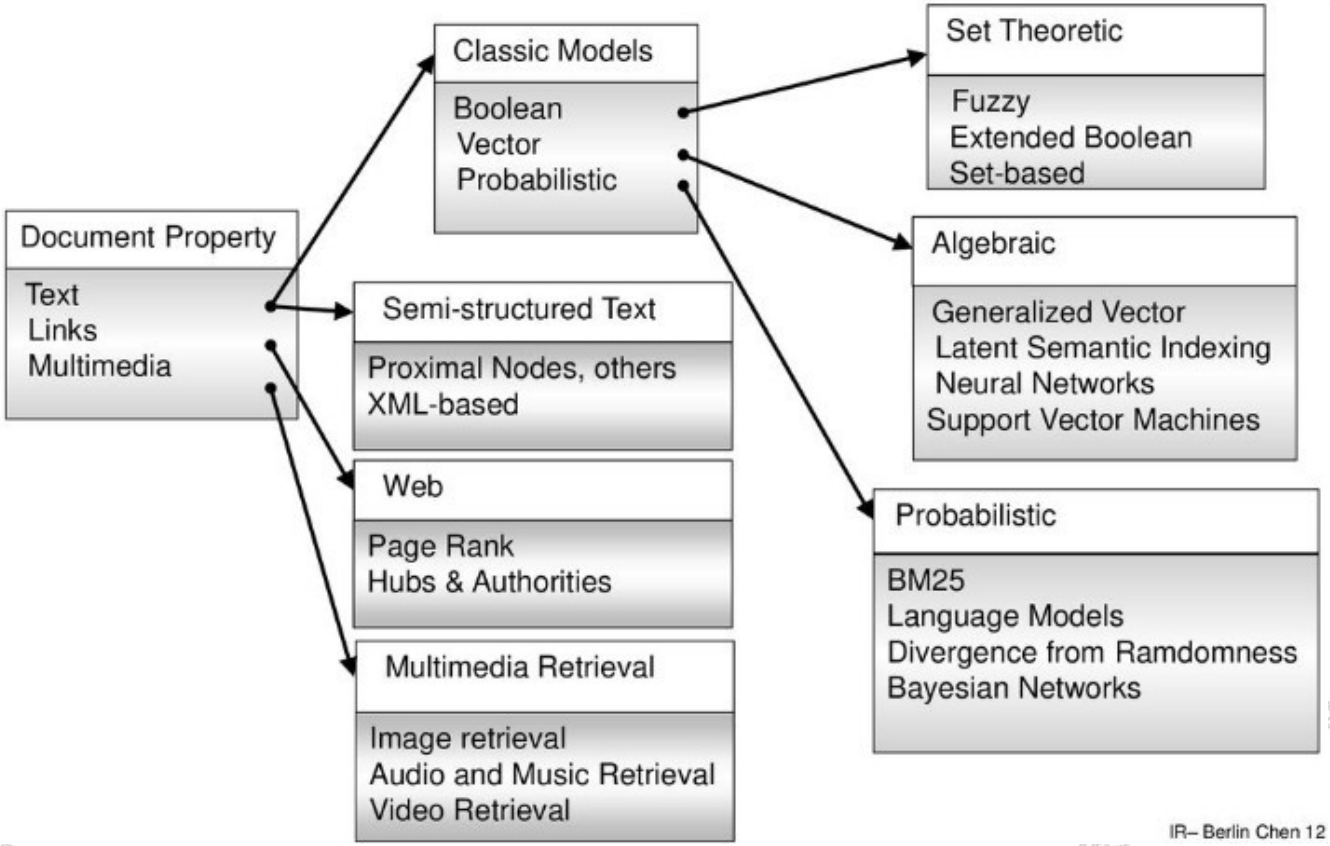
Two Types of Retrieval Tasks

- Adhoc
  - Documents in the collection remain relatively static while new queries are submitted to the system.

- Filtering
  - Queries remain relatively static while new documents get added into the system. Results are not ranked.



Taxonomy of IR – Document Property



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Three Classic Models

Set theoretic .

- Boolean Model
- Fuzzy models
- Extended Boolean Models

Algebraic .

- Vector model
- Generalised vector
- Latent semantic indexing
- Neural Network models

Probabilistic

- Inference network
- Belief network models

Characterization of IR Models

An Information Retrieval Model is a quadruple $[D, Q, F, R(q_i, d_j)]$ where

- D - set of logical representations for the documents in the collection.
- Q - set of queries.
- F - framework for modelling document representations, queries and their relationships.
- $R(q_i, d_j)$ - ranking function which associates a real number with a query $q_i \in Q$ and a document representation $d_j \in D$.

Representation of Documents and Queries

- Boolean model – Documents and queries are represented as sets of index terms.
- Vector model – Documents and queries are represented in t-dimensional space.
- Probabilistic model – Documents and queries are represented based on probability theory.

FRAMEWORK

- Boolean model – Framework is composed of sets of documents and operations on sets.
- Vector model – Framework is composed of a t-dimensional space and linear algebra operations on vectors.
- Probabilistic model – Framework is composed of sets and probability operations and Bayes theorem.

Classic Information Retrieval

- Each document is described by a set of index terms.
- Index terms is a word whose semantics help in remembering the main themes of the document.
- Index terms are used to index and summarise the document contents.
- Index are generally nouns, as semantics of nouns are easier to identify understand.
- Adjective, Adverbs and connectives mainly work as complements, so less useful.
- Properties of index term can be easily measured and are useful for evaluating the potential of the term. Eg: A word that appears in just 5 documents compared to a word that appears in 1000 documents is more useful.
- Distinct index terms have varying relevance and this is captured though numerical weights.

Index Term and its Weight

- t - number of index terms
- k_i - generic index term
- $K = \{k_1, \ldots, k_t\}$ - set of all index terms
- d_j - document
- w_{ij} - weight associated with index term k_i in document d_j. $w_{ij} \geq 0$
- $\vec{d}_j = (w_{1j}, w_{2j}, \ldots, w_{tj})$ - index term vector i.e. weight vector associated with the document d_j
-
- g_i - function that returns the weight associated with the index term k_i
- $g_i(\vec{d}_j) = w_{ij}$
- (Assumption) Index term weights are mutually independent. There is not correlation between the index term weights.

Boolean Model

- Simple model based on set theory and Boolean algebra.
  - Documents are sets of terms.
  - Queries are Boolean expressions on terms.
- Historically the most common model
  - Library OPACs
  - Dialog system
  - Many web search engines
- Queries specified as boolean expressions.
  - Precise semantics
  - Neat formalism
- Terms are either present or absent. $w_{ij} = 1, 0$
- There are three connectives used: and, or, not.

References

1. Modern Information Retrieval, Ricardo Baeza-Yates and Berthier Ribeiro-Neto, Addison-Wesley, 2000. – Chapter 1 and 2.

Thank You!